QUESTION 43: CHEQUE CLEARING DECISION

COBOL Banking Mini-Project – Intermediate

1. AIM

To develop a COBOL program to validate cheque details, including cheque amount, signature match, account status, and date validity. The program accepts the cheque only when the amount is positive and all three validation flags are Y. Otherwise, it rejects the cheque and displays all failed validation reasons.

2. ALGORITHM

1. Start the program.
2. Read the cheque amount.
3. Read the signature-match flag.
4. Read the account-active flag.
5. Read the date-valid flag.
6. Check whether the cheque amount is greater than zero. If not, display the reason: AMOUNT MUST BE POSITIVE.
7. Check whether the signature-match flag is Y. If not, display the reason: SIGNATURE MISMATCH.
8. Check whether the account-active flag is Y. If not, display the reason: ACCOUNT INACTIVE.
9. Check whether the date-valid flag is Y. If not, display the reason: INVALID CHEQUE DATE.
10. If all validations pass, display ACCEPTED.
11. If any validation fails, display REJECTED followed by all applicable failure reasons.
12. Stop the program.

3. COBOL SOURCE CODE

```
IDENTIFICATION DIVISION.
PROGRAM-ID. CHEQUE-CLEARING.

DATA DIVISION.
WORKING-STORAGE SECTION.
01 AMOUNT-TEXT      PIC X(12).
01 CHEQUE-AMOUNT    PIC 9(9)V99.
01 SIGNATURE-MATCH  PIC X.
01 ACCOUNT-ACTIVE   PIC X.
01 DATE-VALID       PIC X.
01 FAILED-CHECK     PIC X VALUE 'N'.

PROCEDURE DIVISION.
  ACCEPT AMOUNT-TEXT
  ACCEPT SIGNATURE-MATCH
  ACCEPT ACCOUNT-ACTIVE
  ACCEPT DATE-VALID
  COMPUTE CHEQUE-AMOUNT =
    FUNCTION NUMVAL(AMOUNT-TEXT)
  IF CHEQUE-AMOUNT <= 0
    DISPLAY "REJECTED"
    DISPLAY "Reason: AMOUNT MUST BE POSITIVE"
    MOVE 'Y' TO FAILED-CHECK
  END-IF

  IF SIGNATURE-MATCH NOT = 'Y'
    IF FAILED-CHECK = 'N'
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      DISPLAY "REJECTED"
    END-IF
    DISPLAY "Reason: SIGNATURE MISMATCH"
    MOVE 'Y' TO FAILED-CHECK
  END-IF

  IF ACCOUNT-ACTIVE NOT = 'Y'
    IF FAILED-CHECK = 'N'
      DISPLAY "REJECTED"
    END-IF
    DISPLAY "Reason: ACCOUNT INACTIVE"
    MOVE 'Y' TO FAILED-CHECK
  END-IF

  IF DATE-VALID NOT = 'Y'
    IF FAILED-CHECK = 'N'
      DISPLAY "REJECTED"
    END-IF
    DISPLAY "Reason: INVALID CHEQUE DATE"
    MOVE 'Y' TO FAILED-CHECK
  END-IF

  IF FAILED-CHECK = 'N'
    DISPLAY "ACCEPTED"
  END-IF

  STOP RUN.
```

4. SAMPLE INPUT

2500.00

Y

N

Y

Input Description:

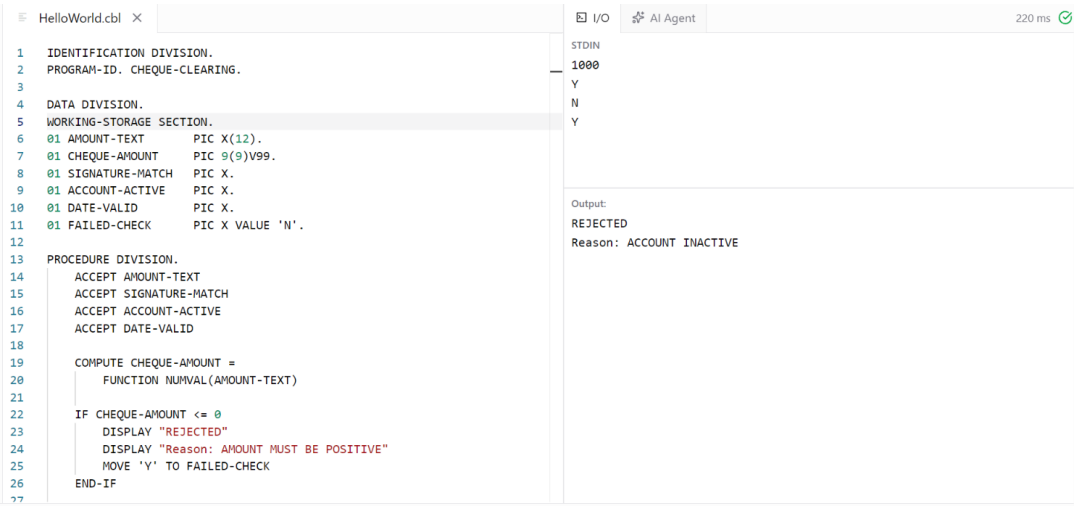
- Cheque Amount: 2500.00
- Signature Match: Y
- Account Active: N
- Date Valid: Y

5. EXPECTED OUTPUT

REJECTED

Reason: ACCOUNT INACTIVE

6. SAMPLE SCREENSHOT



7. RESULT

Thus, the COBOL cheque-clearing program was developed to validate the cheque amount, signature, account status, and date validity. The program successfully identifies failed validation checks and displays the appropriate rejection reasons. For the given sample input, the cheque is rejected because the account is inactive.